SULTAN SHALAHUDDIN

Data Enthusiast

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SUMMARY

Informatics graduate with hands-on experience in data analytics, machine learning, MLOps, and RAG-based applications. Built predictive models, deployable AI solutions, and workflow automation supporting 4,000+ users. Skilled in translating data and operational needs into practical, data-driven solutions.

SKILLS

- **Programming Language:** Python, SQL, HTML, CSS.
- **AI & Data:** Machine Learning, Deep Learning, Computer Vision, NLP, RAG, Time-Series Forecasting, Data Analytics, Data Visualization.
- **Frameworks & Libraries:** scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, Pandas, NumPy, LangChain.
- **MLOps & Deployment:** MLflow, FastAPI, Docker, Prometheus, Grafana, Streamlit, Git/GitHub.
- **BI & Automation:** Power BI, Tableau, Looker Studio, Microsoft Excel, Google Apps Script.
- **Soft Skills:** Analytical Thinking, Problem Solving, Communication, Teamwork, Time Management, Adaptability.

PROJECTS

SmartShopper Assistant: AI Agent with RAG Sep 2026

- Developed an AI shopping assistant agent using **Google ADK**, combining a Product Recommendation tool (embedding-based cosine similarity) with a Common Information tool built on **RAG** (MongoDB Atlas Vector Search + Gemini) for grounded, context-aware responses.
- Designed the retrieval strategy, tool orchestration, and agent prompting to ensure accurate and relevant answers from a custom knowledge base.

House Price Prediction: ML Pipeline & MLOps Deployment Sep 2026

- Built an end-to-end regression pipeline (data cleaning, normalization, feature selection) and trained multiple models including **Random Forest** with varied configurations to predict house sale prices.
- Converted the pipeline into a production-style MLOps project with experiment tracking and model registry via MLflow, served through FastAPI, monitored with Prometheus and Grafana, and containerized with Docker.

SmartSplit Bill AI: Receipt Extraction & Bill-Splitting App Sep 2026

- Built a **computer vision–based** application that extracts itemized data from **receipt images** and automatically calculates bill splits among multiple users, deployed as an interactive Streamlit app.
- Benchmarked multiple vision-language approaches (Gemini API, Groq/Qwen) to select the best-performing model for extraction accuracy and reliability.

CERTIFICATIONS & PROFESSIONAL TRAINING

Dibimbing.id Mar 2026 – Present
Intensive Bootcamp Data Science & AI Machine Learning

- Built end-to-end machine learning projects covering preprocessing, modeling, evaluation, deployment, and monitoring.
- Built AI applications including a RAG-based chatbot, intelligent customer assistant, personalized recommendation system, and computer vision–based bill extraction system using LangChain, PostgreSQL/PgVector, Groq API, and Streamlit.
- Implemented an MLOps pipeline with MLflow, FastAPI, Docker, Prometheus, and Grafana.

Badan Nasional Sertifikasi Profesi (BNSP) Jul 2026

Data Analyst
- Demonstrated competency across seven SKKNI units covering data collection, validation, business intelligence, and analytical reporting.
- Analyzed 51,290 e-commerce records and developed Power BI dashboards for sales performance and RFM customer segmentation.

Google July 2024 – Oct 2024
Project Management Professional Certificate
- Completed a six-course program covering project initiation, planning, execution, Agile/Scrum, and stakeholder management.
- **Core Skills:** Project Planning, Agile/Scrum, Scope Management, Stakeholder Analysis, Project Documentation, and Communication Planning.

MSIB 6 - Startup Campus Feb 2024 – June 2024
Data Science & AI Track
- Ranked 8th best participant out of 696 participants.
- Top 10 Final Project out of 112 teams, presenting our Final Project to Subject Matter Expert.
- Learned about Foundation for Data Science, Data Preprocessing, Machine Learning, Data Visualization, Introduction to Artificial Intelligence

WORK EXPERIENCES

System Administrator Intern Nov 2025 – May 2026

Immigration Office Class I TPI Banda Aceh (MagangHub)

- Developed a workflow automation system using Google Spreadsheet and Google Apps Script to improve service communication efficiency for 4,000+ users.
- Implemented automated notification processes to reduce manual workload and enhance operational efficiency.
- Performed system troubleshooting and maintenance to ensure high system availability and productivity,

Laboratory Assistant Feb 2024 – May 2025

Informatics Department Universitas Syiah Kuala

- Taught Artificial Intelligence, Machine Learning, Computer Vision, and Data Visualization to undergraduate students.
- Translated complex AI and machine learning concepts into simple explanations for non-technical audiences.
- Assisted in developing hands-on lab materials and practical AI exercises.

EDUCATION

Bachelor in Informatics, Syiah Kuala University Aug 2021 – Aug 2025
- GPA: 3.96/4.00
- Best Graduate of the Faculty of Mathematics and Natural Sciences
- Thesis**:** Performance Comparison of LSTM and GRU in Predicting Stock Prices of Energy Sector Companies in the LQ45 Index.
- Relevant Coursework: Artificial Intelligence, Machine Learning, Data Mining, Data Visualization, Advanced Database Systems, Statistics.

ORGANIZATION

Himpunan Mahasiswa Informatika USK Feb 2023 – Jan 2024

Staff of Student Welfare Management
- Involved in 9+ campus event in various divisions such as Food and Beverages, Administration, Public Relations, and Event division.
- Created a Guest Inspiring Talk event that reached 70+ audiences.
- Involved in planning innovations and creating new programs in collaboration with the department head, deputy head and secretary department.